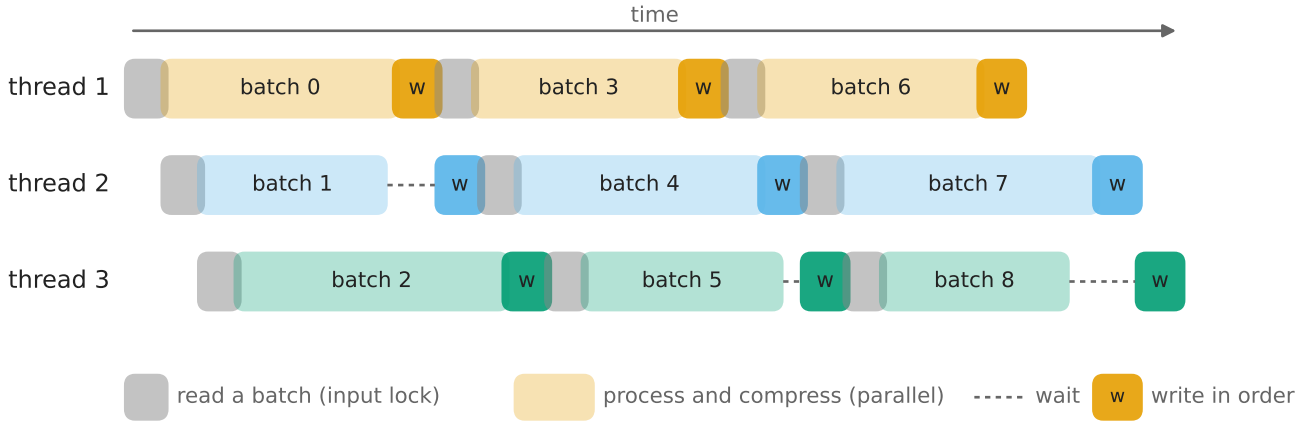
time
thread 1
batch 0
w
batch 3
w
batch 6
w
thread 2
batch 1
w
batch 4
w
batch 7
w
thread 3
batch 2
w
batch 5
w
batch 8
w
read a batch (input lock)
process and compress (parallel)
wait
w
write in order